Multimodal Diffusion-Based Depth Estimation Framework with Wi-Fi and Vision

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Depth estimation is essential for applications such as autonomous driving, robotic navigation, and augmented reality. While LiDAR-based and mmWave-based approaches offer high accuracy, they come with substantial deployment costs, whereas monocular vision methods often struggle with precision. To overcome these challenges, we propose *WiViD*, a novel diffusion-based depth estimation framework that integrates commercial Wi-Fi and visual data. By leveraging the iterative refinement capabilities of diffusion models, *WiViD* achieves high-precision depth predictions. *WiViD* features two key encoding modules: the Complex-Valued CSI Encoder (CCE), which extracts rich spatio-temporal features from Wi-Fi signals, and the Residual Image Encoder (RIE), which processes visual data. Additionally, we introduce the Multi-Modal Alignment Bidirectional Cross-Attention (MABA) mechanism to enhance cross-modal feature integration. Real-world experiments demonstrate that *WiViD* significantly outperforms state-of-the-art monocular depth estimation methods, reducing Absolute Relative Error (ARE) and Square Relative Error (SRE) by 33.3% and 9.5%, respectively, highlighting its superior accuracy and robustness.

Additional Key Words and Phrases: Depth estimation, Wireless sensing, Diffusion model, Multimodal fusion, Wi-Fi CSI

1 Introduction

Depth estimation is fundamental to 3D environmental perception, which is a key capability for applications such as autonomous driving, robotic navigation, and augmented reality [1]. Over the past decades, extensive research has been dedicated to advancing this field.

Depth estimation methods are generally categorized into active and passive approaches. Active methods, such as mmWave radar [23] and LiDAR [21], determine depth by emitting millimeter waves or lasers and analyzing their reflections. While highly precise in localization [36], their reliance on costly hardware hinders large-scale deployment. In contrast, passive methods, typically based on a single image, are more cost-effective but sacrifice accuracy. Monocular vision, for instance, estimates depth from RGB images, providing a low-cost and portable solution but remaining sensitive to lighting and texture variations. To improve performance, recent studies have explored multi-modal approaches that integrate RGB with infrared sensing [2], LiDAR [19], or mmWave [27]. However, these methods introduce substantial computational overhead while still depending on

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expensive hardware. Consequently, achieving accurate, robust, and cost-effective depth estimation remains a major challenge.

Recently, with the rapid development of wireless communication technology, Wi-Fi-based localization [5, 9, 15] and sensing [6, 34, 39] have become key research areas due to their ubiquitous availability, low cost, and ease of deployment. This Wi-Fi-based sensing solution eliminates the need for expensive hardware while offering pervasive coverage.

Considering the above situation, we identify an opportunity to achieve depth estimation by integrating two complementary modalities: Wi-Fi and vision. Wi-Fi Channel State Information (CSI) [33, 35] captures three-dimensional spatial data by detecting variations in signal reflection caused by environmental objects, making it highly sensitive to changes in the signal propagation path (i.e., the radial direction). In contrast, vision-based methods excel at detecting changes in the pixel plane (i.e., the tangential direction), providing detailed spatial information with high pixel-level precision. By fusing these two modalities, we can create a multi-modal solution that significantly enhances both the accuracy and robustness of depth estimation, offering more precise and reliable 3D perception for advanced applications.

However, implementing this intuition in a practical system faces two main challenges.

Challenge 1: Designing a High-Precision Wi-Fi and Vision multi-modal Depth Estimation Framework. Previous approaches that combine Wi-Fi and vision have primarily focused on classification tasks, such as human activity recognition. In contrast, depth estimation is a regression task that not only requires higher precision but also demands continuous, high-resolution distance data. Unlike classification, which deals with discrete labels, depth estimation involves continuous variables, making traditional data fusion techniques less effective in this context.

Challenge 2: Extracting Precise Spatial Information from Complex-Valued Wi-Fi CSI Signals. Wi-Fi CSI signals exhibit complex domain characteristics, including both amplitude and phase information. Existing real-domain processing methods are inadequate for handling the intricate spatio-temporal coupling inherent in the complex domain. Due to multipath propagation, the blending of CSI signal information across both time and space complicates feature extraction. As a result, traditional approaches fail to fully exploit the rich spatio-temporal information embedded in complex-valued CSI signals. Consequently, developing a neural network capable of operating in the complex domain to effectively extract information from CSI signals for depth reconstruction has become a key research challenge.

To address these challenges, we propose *WiViD*, a multi-modal depth estimation system based on a generative diffusion model. By leveraging iterative refinement, diffusion models excel at generating high-quality, detailed predictions. *WiViD* features two specialized network structures for processing different data modalities: the Complex-Valued CSI Encoder (CCE) for encoding complex-valued Wi-Fi CSI and the Residual Image Encoder (RIE) for extracting features from RGB images. To effectively integrate these modalities, we introduce the Multi-Modal Alignment Bidirectional Cross-Attention (MABA) mechanism, which employs cross-attention to exchange and refine information between CCE and RIE before incorporating it into the control condition. To further accelerate the sampling process, we propose an Accelerated Denoising Sampling method, which optimizes the generative diffusion process. Within the U-Net [25] Denoising Module, the control condition guides the denoising process, enabling precise depth estimation.

We implement *WiViD* and conduct extensive real-world experiments, comparing it with four state-of-the-art monocular depth estimation methods: DORN[8], VA-DepthNet[16], VGGT, and Depth Anything V2. Results show *WiViD* achieves an ARE of 0.034, outperforming DORN (0.153), VA-DepthNet (0.059), VGGT[31] (0.062), and Depth Anything V2[32] (0.051), while additional experiments validate the effectiveness of its multi-modal fusion mechanism.

Our contributions are summarized as follows:

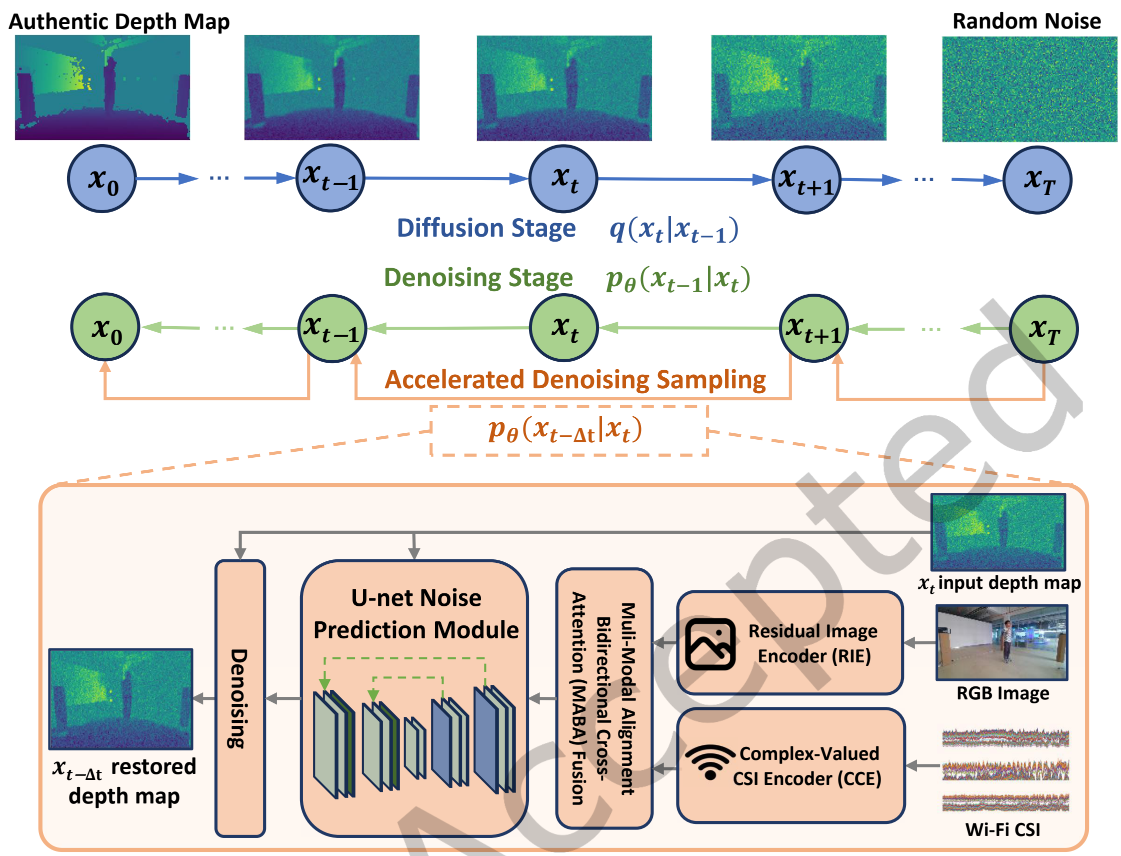


Fig. 1. System Overview of *WiViD*

• We propose *WiViD*, the first depth estimation system incorporating an innovative Wi-Fi and vision multi-modal diffusion model. *WiViD* achieves excellent experimental results, representing a promising step in exploring depth estimation by fusing Wi-Fi and vision, while notably enhancing robustness against challenging indoor lighting variations.
• Our proposed Complex-Valued CSI Encoder (CCE) offers unique advantages in handling spatio-temporal complex-valued signals and can be directly applied to other wireless sensing applications (e.g., human gesture recognition, fall detection) to enhance their feature extraction capabilities.
• Our proposed Multi-modal Alignment Bidirectional Cross-Attention (MABA) mechanism is a pioneering approach to Wi-Fi-vision fusion within a diffusion framework. The MABA mechanism facilitates the exchange and refinement of information between the CCE and the RIE through cross-attention, enabling enhanced multi-modal perception. This approach can be extended to other sensing modalities, establishing a new paradigm for multi-modal fusion in perception tasks.

2 System Overview

WiViD is a Wi-Fi and vision depth estimation system that utilizes multi-modal diffusion. As shown in Fig. 1, *WiViD* extracts features from Wi-Fi CSI signals and RGB images, encoding them as control conditions to guide the diffusion model in generating accurate depth maps. To effectively combine these modalities, *WiViD* incorporates a fusion mechanism that enables bidirectional interaction between Wi-Fi and vision features, allowing the model to refine information and enhance depth estimation accuracy.

Similar to DDPM [12], our diffusion model comprises two stages: forward diffusion and reverse denoising. In the forward diffusion stage, Gaussian noise is progressively added to the depth map, converting it into a noise distribution. During the reverse denoising stage, *WiViD* uses the U-net Noise Prediction Module to estimate the noise in the $\boldsymbol{x}_t$ input depth map. It then iteratively removes the predicted noise, reconstructing the depth map step by step until $\boldsymbol{x}_0$, the final noise-free output. Additionally, we propose the Accelerated Denoising Sampling strategy, which speeds up the reverse denoising process while maintaining prediction accuracy. The training process includes both stages, while the prediction process only requires the reverse denoising stage.

The architecture of *WiViD* consists of three key components: the Complex-Valued CSI Encoder (CCE), the Residual Image Encoder (RIE), and the U-net Noise Prediction Module. We employ the Multi-Modal Alignment Bidirectional Cross-Attention (MABA) mechanism for the fusion of different modalities.

When performing depth estimation tasks, the algorithm samples random Gaussian noise and gradually transforms it into a depth map. The CCE and RIE extract and encode features from the input Wi-Fi CSI and RGB images into high-dimensional vectors. These vectors are fused using MABA, which facilitates the interaction between Wi-Fi and vision features to form a comprehensive representation. This representation serves as a control condition during the denoising stage, embedding spatial information from both modalities to enhance depth estimation. In each block of the U-net Noise Prediction Module, this control condition is incorporated into the predicted intermediate representation, guiding the denoising process to improve depth estimation accuracy.

3 *WiViD* Design

WiViD is designed for depth estimation and proposes a deep learning strategy based on a diffusion model. It addresses two major challenges: 1) extracting and encoding features from RGB images and Wi-Fi CSI signals, particularly Wi-Fi CSI; and 2) fusing the encoded information from these two modalities to guide the denoising stage of the diffusion model for depth estimation.

To address these challenges, we introduce the RGB Image Encoder (RIE) and the Complex-Valued CSI Encoder (CCE) into the diffusion model. And we employ the Multi-Modal Alignment Bidirectional Cross-Attention (MABA) mechanism for multi-modal fusion.

3.1 Diffusion for Depth Estimation

The diffusion model for *WiViD* consists of two stages: the diffusion stage and the denoising stage.

During the diffusion process, the original depth map $\boldsymbol{x}_0$ undergoes a gradual degradation as random Gaussian noise is progressively introduced. This process gradually obliterates the original information until the depth map is transformed into pure noise $\boldsymbol{x}_T$. The diffusion process consists of T steps, where each step $t \in \{1, ..., T\}$ is characterized by a noise parameter β_t, which determines the magnitude of the introduced noise. Furthermore, we define $\alpha_t = 1 - \beta_t$ as the noise retention factor at each step and introduce $\bar{\alpha}_t = \prod_{i=0}^{t} \alpha_i$ to quantify the cumulative preservation of the depth map's structural integrity up to step t. The diffusion process of the depth map is formulated as follows:

$$q(\boldsymbol{x}_t|\boldsymbol{x}_0) \sim \mathcal{N}\left(\boldsymbol{x}_t; \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right), \tag{1}$$

which is equivalent to:

$$\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \tag{2}$$

where $\boldsymbol{\epsilon}$ represents the standard Gaussian noise.

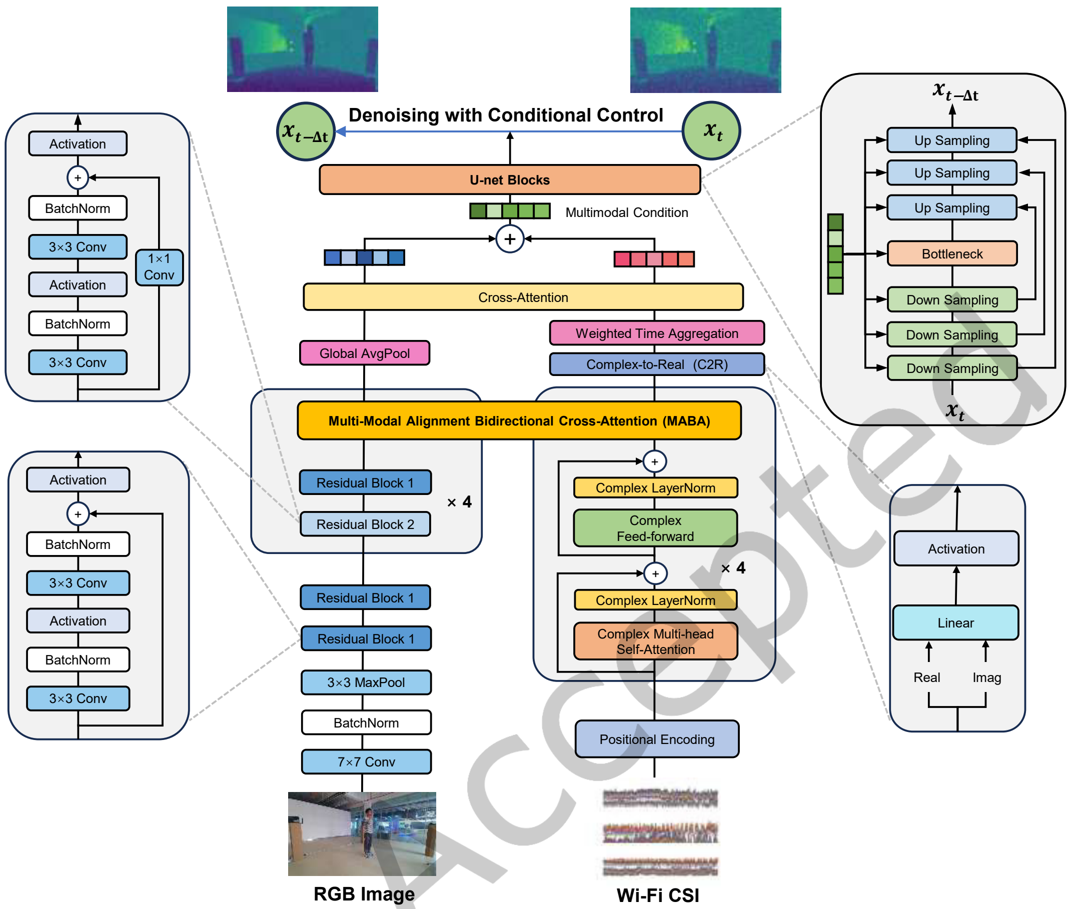


Fig. 2. Design of Denoising with Conditional Control

During the denoising stage, a deep neural network $\boldsymbol{\epsilon}_\theta$ is trained to estimate the noise component in the corrupted depth map $\boldsymbol{x}_t$, conditioned on the step number t, RGB image $\boldsymbol{M}$ and Wi-Fi CSI $\boldsymbol{H}$. Unlike conventional diffusion models, *WiViD* employs an accelerated denoising strategy inspired by Denoising Diffusion Implicit Models (DDIM) [28] to enhance efficiency in depth map generation.

To formulate this accelerated sampling process, consider two arbitrary steps k and l ($k, l \in \{1, ..., T\}$, $k < l$). The transition from $\boldsymbol{x}_l$ to $\boldsymbol{x}_k$ is given by:

$$\boldsymbol{x}_k = \sqrt{\bar{\alpha}_k}\tilde{\boldsymbol{x}}_0 + \sqrt{1 - \bar{\alpha}_k}\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_l, l, \boldsymbol{M}, \boldsymbol{H}), \tag{3}$$

where $\tilde{\boldsymbol{x}}_0$ represents the estimated authentic depth map $\boldsymbol{x}_0$, derived from $\boldsymbol{x}_l$ as follows:

$$\tilde{\boldsymbol{x}}_0 = \frac{\boldsymbol{x}_l - \sqrt{1 - \bar{\alpha}_l} \cdot \boldsymbol{\epsilon}_\theta(\boldsymbol{x}_l, l, \boldsymbol{M}, \boldsymbol{H})}{\sqrt{\bar{\alpha}_l}}. \tag{4}$$

Substituting Eq. 4 into Eq. 3, the direct transition from x_l to $x_k$is expressed as:

$$\boldsymbol{x}_k = \frac{\sqrt{\bar{\alpha}_k}}{\sqrt{\bar{\alpha}_l}}\boldsymbol{x}_l + \left(\sqrt{1-\bar{\alpha}_k} - \frac{\sqrt{\bar{\alpha}_k(1-\bar{\alpha}_l)}}{\sqrt{\bar{\alpha}_l}}\right)\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_l, l, \boldsymbol{M}, \boldsymbol{H}). \tag{5}$$

To implement this accelerated sampling scheme, we introduce two parameters: T_a and Δt. The step size Δt determines the interval between selected steps, while the total number of sampling steps T_a is given by:

$$T_a = \left\lceil \frac{T}{\Delta t} \right\rceil, \tag{6}$$

where T is the original total number of steps, and the ceil function ensures an integer result. This formulation enables approximately evenly spaced steps, balancing computational efficiency and generation quality. Thus, the process of accelerated denoising sampling is described as follows:

$$p_\theta(\boldsymbol{x}_{t-\Delta t}|\boldsymbol{x}_t) \sim \mathcal{N}\left(\boldsymbol{x}_{t-\Delta t}; \sqrt{1-\bar{\alpha}_{t-\Delta t}}\boldsymbol{x}_t, (\sqrt{1-\bar{\alpha}_{t-\Delta t}} - \frac{\sqrt{\bar{\alpha}_{t-\Delta t}(1-\bar{\alpha}_t)}}{\sqrt{\bar{\alpha}_t}})^2\mathbf{I}\right), \tag{7}$$

which is equivalent to:

$$\boldsymbol{x}_{t-\Delta t} = \frac{\sqrt{\bar{\alpha}_{t-\Delta t}}}{\sqrt{\bar{\alpha}_t}}\boldsymbol{x}_t + \left(\sqrt{1-\bar{\alpha}_{t-\Delta t}} - \frac{\sqrt{\bar{\alpha}_{t-\Delta t}(1-\bar{\alpha}_t)}}{\sqrt{\bar{\alpha}_t}}\right)\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t, \boldsymbol{M}, \boldsymbol{H}). \tag{8}$$

Algorithm 1 *WiViD* Training Algorithm

Input: Dataset following $\boldsymbol{x} \sim q(\boldsymbol{x})$, image $\boldsymbol{M}$ and CSI $\boldsymbol{H}$
Output: Noise prediction model $\boldsymbol{\epsilon}_\theta$

```
repeat
    Sample x_0 ~ q(x_0), M and H from dataset
    Sample t ~ Uniform({1, ..., T})
    Sample ε ~ N(0, I)
    Compute x_t = √(ᾱ_t) x_0 + √(1 − ᾱ_t) ε
    Take gradient descent step on
        ∇_θ ||ε − ε_θ(x_t, t, M, H)||^2
until converged
```

The training process of *WiViD* comprises a diffusion stage and a denoising stage, as illustrated in Algorithm 1. Noise is added to the authentic depth map $\boldsymbol{x}_0$, with the noise level determined by t, randomly selected from $\{1, ..., T\}$. Simultaneously, the image $\boldsymbol{M}$ and CSI $\boldsymbol{H}$ serve as control conditions. The model $\boldsymbol{\epsilon}_\theta$ then predicts the added noise to reconstruct $\boldsymbol{x}_0$ from the corrupted data $\boldsymbol{x}_t$. Training optimizes $\boldsymbol{\epsilon}_\theta$ to minimize the discrepancy between the predicted and actual noise.

The sampling process in *WiViD* involves only the denoising stage, as outlined in Algorithm 2. *WiViD* employs the accelerated sampling strategy to efficiently perform the denoising process. Starting from the final step T, Gaussian noise $\boldsymbol{x}_T$ is randomly generated, representing the starting point for the reverse process. *WiViD* progresses through T_a steps to remove noise, along with the corresponding RGB image $\boldsymbol{M}$ and Wi-Fi CSI $\boldsymbol{H}$ are provided as conditions to guide the denoising process. At each step, the model $\boldsymbol{\epsilon}_\theta$ predicts the noise during that step and refines the estimate of the depth map. To reduce the number of iterations, the strategy selects intermediate steps with a larger step size Δt, ensuring that the reconstruction quality is preserved. This process continues until the noise is sufficiently removed, ultimately recovering the authentic depth map $\boldsymbol{x}_0$.

Algorithm 2 *WiViD* Sampling Algorithm

Input: Noise prediction model $\boldsymbol{\epsilon}_\theta$, image $\boldsymbol{M}$ and CSI $\boldsymbol{H}$
Output: Depth map $\boldsymbol{x}_0$

Sample $\boldsymbol{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
Let $t = T$
while $t > 0$ **do**
  if $t - \Delta t > 0$ **then**
    Compute $\boldsymbol{x}_{t-\Delta t} = \frac{\sqrt{\bar{\alpha}_{t-\Delta t}}}{\sqrt{\bar{\alpha}_t}}\boldsymbol{x}_t + \left(\sqrt{1-\bar{\alpha}_{t-\Delta t}} - \frac{\sqrt{\bar{\alpha}_{t-\Delta t}(1-\bar{\alpha}_t)}}{\sqrt{\bar{\alpha}_t}}\right)\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t, \boldsymbol{M}, \boldsymbol{H})$
  else
    Compute $\boldsymbol{x}_0 = \frac{\sqrt{\bar{\alpha}_0}}{\sqrt{\bar{\alpha}_t}}\boldsymbol{x}_t + \left(\sqrt{1-\bar{\alpha}_0} - \frac{\sqrt{\bar{\alpha}_0(1-\bar{\alpha}_t)}}{\sqrt{\bar{\alpha}_t}}\right)\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t, \boldsymbol{M}, \boldsymbol{H})$
  end if
  Let $t = t - \Delta t$
end while
return $\boldsymbol{x}_0$

For depth images of *WiViD*, depth can be represented as $\mathbf{d} \in \mathbb{R}^{1\times M\times N}$, where M and N denote the height and width of the depth map. Unlike RGB images with three channels, depth maps have only one channel. The depth map is normalized to the range $[-1, 1]$ to match the Gaussian noise mean and variance in the diffusion model. So that the diffusion and denoising stages are applied to these normalized depth maps.

3.2 Residual Image Encoder (RIE)

To fully leverage the information in RGB images, we design the RIE module based on ResNet [11] for feature extraction and encoding. This module encodes the image $\boldsymbol{M}$ into a vector for depth estimation and consists of three components: Initial Feature Module (IFM), Deep Residual Module (DRM), and Global Feature Module (GFM).

Initial Feature Module. The IFM module comprises a 7x7 convolutional layer, a Batch Normalization layer, and a 3x3 MaxPooling layer. It is intended for the initial extraction of features from the input image. Given an input data $\boldsymbol{X}$, the IFM module is formulated as follows:

$$\mathrm{IFM}(\boldsymbol{X}) = \mathrm{MaxPool}(\sigma(\mathrm{BN}(\mathrm{Conv}_{7x7}(\boldsymbol{X})))), \tag{9}$$

where $\mathrm{IFM}(\cdot)$ represents the process of the preprocessing module, and $\sigma(\cdot)$ represents the activation function.

Deep Residual Module. The DRM is a crucial component of RIE, enabling the effective training of very deep networks. It mitigates the vanishing gradient problem and enables the model to learn residual mappings instead of direct mappings, thereby facilitating the training of deeper networks.

The DRM primarily consists of two types of Residual Blocks. Each Residual Block comprises a series of convolutional layers, BatchNorm layers, and activation functions, described as follows:

$$\mathrm{RB}_1(\boldsymbol{X}) = \sigma(\mathrm{BN}(\mathrm{Conv}_{3x3}(\sigma(BN(\mathrm{Conv}_{3x3}(\boldsymbol{X}))) + \boldsymbol{X}), \tag{10}$$

$$\mathrm{RB}_2(\boldsymbol{X}) = \sigma(\mathrm{BN}(\mathrm{Conv}_{3x3}(\sigma(BN(\mathrm{Conv}_{3x3}(\boldsymbol{X}))) + \mathrm{Conv}_{1X1}(\boldsymbol{X})), \tag{11}$$

where $\mathrm{RB}_1(\cdot)$ and $\mathrm{RB}_2(\cdot)$ respectively represent residual blocks of two different structures.

In the DRM, the data first passes through two RB_1 structures, followed by four composite structures consisting of RB_2 and RB_1, described as follows:

$$\mathrm{DRM}(\boldsymbol{X}) = (\mathrm{RB}_2 \circ \mathrm{RB}_1)^4(\mathrm{RB}_1^2(\mathbf{X})), \tag{12}$$

where $\mathrm{DRM}(\cdot)$ denotes the process of the DRM.

Global Feature Module. The GFM extracts global feature representations by performing average pooling on each channel of the input feature matrix. Specifically, it independently computes the spatial average of each channel, reducing spatial dimensions while retaining channel-wise information.

$$\mathrm{GFM}(\boldsymbol{X}) = \mathrm{AvgPool}(\boldsymbol{X}), \tag{13}$$

where $\mathrm{GFM}(\cdot)$ represents the process of the GFM.

Therefore, RIE can be represented by the following formula:

$$\mathrm{RIE}(\boldsymbol{M}) = \mathrm{GFM}(\mathrm{DRM}(\mathrm{IFM}(\boldsymbol{M}))), \tag{14}$$

where $\boldsymbol{M}$ denotes the input image.

3.3 Complex-Valued CSI Encoder (CCE)

To effectively encode Wi-Fi CSI data and extract its spatio-temporal features, we design a Multi-Head Transformer Encoder that supports complex-valued calculation. This encoder not only preserves the authentic information of the CSI signal but also captures the interactions and dependencies between signals through the Attention mechanism.

Complex-Valued Multi-Head Attention. In this Transformer Encoder, we extend Attention to the complex domain. For given $\boldsymbol{X} \in \mathbb{C}^{L\times d_{\mathrm{in}}}$, firstly it will perform linear projection:

$$\boldsymbol{Q} = \boldsymbol{X}\boldsymbol{W}_Q, \quad \boldsymbol{K} = \boldsymbol{X}\boldsymbol{W}_K, \quad \boldsymbol{V} = \boldsymbol{X}\boldsymbol{W}_V, \tag{15}$$

then we get $\boldsymbol{Q} \in \mathbb{C}^{L\times d_{\mathrm{Q}}}$, $\boldsymbol{K} \in \mathbb{C}^{L\times d_{\mathrm{K}}}$ and $\boldsymbol{V} \in \mathbb{C}^{L\times d_{\mathrm{V}}}$, in which the multiplication follows the principles of complex multiplication, and $\boldsymbol{W}_Q \in \mathbb{C}^{d_{\mathrm{in}}\times d_Q}$, $\boldsymbol{W}_K \in \mathbb{C}^{d_{\mathrm{in}}\times d_K}$ and $\boldsymbol{W}_V \in \mathbb{C}^{d_{\mathrm{in}}\times d_v}$ are projection parameters. Softmax and Attention mechanism are also extended in complex domain, and can be defined as follows:

$$\mathrm{softmax}_{\mathrm{complex}}(z) = \frac{\mathrm{e}^{|z|}\mathrm{e}^{j\angle(z)}}{\sum_{i=1}^{n}\mathrm{e}^{|z_i|}}, \tag{16}$$

$$\mathrm{Attention}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = \mathrm{softmax}_{\mathrm{complex}}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{d_K}}\right)\boldsymbol{V}, \tag{17}$$

where j is the imaginary unit. Furthermore, we employ Multi-Head Attention with h heads, defined as follows:

$$\mathrm{MultiHead}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = (a_1, a_2, \ldots, a_h)^T\boldsymbol{W}_O, \tag{18}$$

where $\boldsymbol{W}_O \in \mathbb{C}^{hd_V\times d_O}$ is the final projection parameter, and $a_i = \mathrm{Attention}(\boldsymbol{X}\boldsymbol{W}_Q^i, \boldsymbol{X}\boldsymbol{W}_K^i, \boldsymbol{X}\boldsymbol{W}_V^i)$.

Complex feed-forward module. The complex feed-forward module contains linear transformations and activation functions SiLU, which support calculations in the complex domain, and can be defined as follows:

$$\boldsymbol{w}\boldsymbol{x}+\boldsymbol{b} = \begin{bmatrix}\Re(\boldsymbol{w}\boldsymbol{x}+\boldsymbol{b})\\ \Im(\boldsymbol{w}\boldsymbol{x}+\boldsymbol{b})\end{bmatrix} = \begin{bmatrix} w_r & -w_i \\ w_i & w_r\end{bmatrix}\begin{bmatrix}x_r\\ x_i\end{bmatrix} + \begin{bmatrix}b_r\\ b_i\end{bmatrix}, \tag{19}$$

$$\mathrm{SiLU}_{\mathrm{complex}}(x) = \mathrm{SiLU}(x_r) + \mathrm{j}\cdot\mathrm{SiLU}(x_i), \tag{20}$$

The Complex Transformer Encoder block consists of Complex Multi-Head Attention and Complex Feed-Forward modules. The Complex Transformer Encoder is structured with 4 of these blocks.

Complex Layer Normalization. We design the Complex Layer Normalization. For given $\boldsymbol{X} \in \mathbb{C}^{L\times d_{\mathrm{in}}}$, firstly the mean and variance are computed:

$$\mu = mean(\boldsymbol{X}), \tag{21}$$

$$\sigma_{real}^2 = var(\boldsymbol{X}_{real}), \tag{22}$$

$$\sigma_{imag}^2 = var(\boldsymbol{X}_{imag}), \tag{23}$$

$$\sigma^2 = \sigma^2_{real} + \sigma^2_{imag}, \tag{24}$$

where μ represents the mean and σ^2 represents the variance, and σ^2_{real} and σ^2_{imag} are the variance in real and imaginary parts of $\boldsymbol{X}$ respectively.

The Complex Layer Normalization includes two sets of learnable parameters, $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$, serving as scaling and bias terms, designed as follows:

$$\boldsymbol{\gamma} = \begin{pmatrix} \gamma_{rr} & \gamma_{ri} \\ \gamma_{ri} & \gamma_{ii} \end{pmatrix}, \tag{25}$$

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_{real} \\ \beta_{imag} \end{pmatrix}, \tag{26}$$

where $\boldsymbol{\gamma}$ is a 2×2 transformation matrix: γ_{rr} and γ_{ii} act as scaling factors for the real and imaginary parts, respectively, while γ_{ri} models the cross-term interaction between real and imaginary components. $\boldsymbol{\beta}$ is a 2-dimensional bias vector, where β_{real} and β_{imag} correspond to the bias terms applied to the real and imaginary parts of the normalized features.

The Complex Layer Normalization is described as follows:

$$CLN(\boldsymbol{X}) = \boldsymbol{\gamma} \frac{\boldsymbol{X} - \mu}{\sqrt{\sigma^2 + \epsilon}} + \boldsymbol{\beta}, \tag{27}$$

where $\epsilon = 10^{-6}$ to avoid division by zero error.

Complex-to-Real Transformation Layer (C2R). Subsequently, we design a Complex-to-Real Transformation Layer following the Transformer Encoder. This layer converts complex features to real features. C2R is defined as follows:

$$\text{C2R}(\boldsymbol{X}) = \sigma(\text{Linear}_1(\Re(\boldsymbol{X})) + \text{Linear}_2(\Im(\boldsymbol{X}))). \tag{28}$$

Following the Transformer Encoder and C2R, the Weighted Time Aggregation (WTA) mechanism is employed to integrate temporal features from CSI data. This mechanism adaptively assigns weights to CSI features at different time points within a short time window, aggregating them into a weighted representation of the Wi-Fi signal distribution at a given moment. By dynamically capturing the relative importance of each temporal feature, WTA effectively extracts and compresses time-dependent information, yielding a compact representation that aligns with single-frame image features. The WTA is described as follows:

$$\text{WTA}(\boldsymbol{X}) = \sigma\left(W_1 \sum_{d=1}^{D} \gamma_d \boldsymbol{x}_d + b_d\right), \tag{29}$$

$$\gamma_d = \text{softmax}\left(W_2 \cdot \sigma(W_3 \boldsymbol{X} + b_1) + b_2\right), \tag{30}$$

where WTA$(\cdot)$ denotes the process of the Weighted Time Aggregation mechanism, and $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_D]$ is the input sequence of temporal feature vectors in which D denotes the dimension of time in $\boldsymbol{X}$. The weight vector γ_d is obtained through softmax normalization. The parameters W_1, W_2, W_3 and b_1, b_2, b_d are learnable weights and biases.

In summary, the process of CCE can be described as follows:

$$\text{CCE}(\boldsymbol{H}) = \text{WTA}(\text{C2R}(\text{CTransEnc}(\boldsymbol{H}))), \tag{31}$$

where CTransEnc represents the Complex Transformer Encoder, $\boldsymbol{H}$ is the Wi-Fi CSI matrix.

3.4 Multi-modal Alignment Bidirectional Cross-Attention (MABA)

To address heterogeneous fusion between RGB image features and Wi-Fi CSI signals, we propose a multi-modal alignment bidirectional cross-attention (MABA) mechanism, establishing bidirectional associations through unified sequence modeling. Unlike existing cross-attention mechanisms that operate on local patches or time sequences, we treat the entire spatial domain of RGB images and the time-frequency domain of Wi-FI CSI as holistic sequences, enabling global correlation modeling between the two modalities without prior structural constraints.

Given RGB features $\boldsymbol{X}_M \in \mathbb{R}^{C\times H\times W}$ (C: channel, H: height, W: width) and CSI features $\boldsymbol{X}_H \in \mathbb{R}^{T\times S\times 2}$ (T: time, S: subcarrier; the last dimension represents real and imaginary parts, so $\mathbb{R}^{T\times S\times 2} \equiv \mathbb{C}^{T\times S}$), we first convert both into sequence representations by flattening their spatial and time-frequency dimensions, resulting in $\boldsymbol{X}_M^{seq} \in \mathbb{R}^{N_M\times C}$ ($N_M = H\times W$) and $\boldsymbol{X}_H^{seq} \in \mathbb{R}^{N_H\times 2}$ ($N_H = T\times S$). This unifies the two modalities into observation sequences, each element corresponding to a scene component captured by either the RGB image or the Wi-Fi CSI signals.

Then, we project both modalities into high-dimensional features using convolutional layers, described as follows:

$$\boldsymbol{X}_M^{proj} = \mathrm{Conv}(\boldsymbol{X}_M^{seq}) \in \mathbb{R}^{N_M\times d_{unify}}, \quad \boldsymbol{X}_H^{proj} = \mathrm{Conv}(\boldsymbol{X}_H^{seq}) \in \mathbb{R}^{N_H\times d_{unify}}, \tag{32}$$

where d_{unify} represents the unified feature dimensions.

And we compute the query, key, and value matrices for the image feature representations $\boldsymbol{X}_M^{proj}$ and the Wi-Fi CSI feature representation $\boldsymbol{X}_H^{proj}$:

$$\boldsymbol{Q}_M = \boldsymbol{X}_M^{proj}\boldsymbol{W}_{QM}, \quad \boldsymbol{K}_M = \boldsymbol{X}_M^{proj}\boldsymbol{W}_{KM}, \quad \boldsymbol{V}_M = \boldsymbol{X}_M^{proj}\boldsymbol{W}_{VM}, \tag{33}$$

$$\boldsymbol{Q}_H = \boldsymbol{X}_H^{proj}\boldsymbol{W}_{QH}, \quad \boldsymbol{K}_H = \boldsymbol{X}_H^{proj}\boldsymbol{W}_{KH}, \quad \boldsymbol{V}_H = \boldsymbol{X}_H^{proj}\boldsymbol{W}_{VH}. \tag{34}$$

Next, we calculate two Cross-Attention mechanisms separately using Multi-head Attention, so that each modality pays attention to the other modality:

$$\boldsymbol{X}_M^{proj\prime} = \mathrm{CrossAttention}(\boldsymbol{X}_M^{proj}, \boldsymbol{X}_H^{proj}) = \mathrm{MultiHead}(\boldsymbol{Q}_M, \boldsymbol{K}_H, \boldsymbol{V}_H), \tag{35}$$

$$\boldsymbol{X}_H^{proj\prime} = \mathrm{CrossAttention}(\boldsymbol{X}_H^{proj}, \boldsymbol{X}_M^{proj}) = \mathrm{MultiHead}(\boldsymbol{Q}_H, \boldsymbol{K}_M, \boldsymbol{V}_M). \tag{36}$$

After obtaining the cross-attention outputs, we recover the high-dimensional attention features back to their original modal dimensions using convolutional layers, ensuring consistency with the input feature dimensions for subsequent processing:

$$\boldsymbol{X}_M^{seq\prime} = \mathrm{Conv}(\boldsymbol{X}_M^{proj\prime}) \in \mathbb{R}^{N_M\times C}, \boldsymbol{X}_H^{seq\prime} = \mathrm{Conv}(\boldsymbol{X}_H^{proj\prime}) \in \mathbb{R}^{N_H\times 2}. \tag{37}$$

Following the dimensionality recovery via convolution, we rearrange the flattened sequences back to their original spatial and time-frequency structures to get $\boldsymbol{X}'_M \in \mathbb{R}^{C\times H\times W}$ and $\boldsymbol{X}'_H \in \mathbb{R}^{T\times S\times 2}$.

Finally, through residual connections, we obtain all the calculation processes of MABA:

$$\mathrm{MABA}(\boldsymbol{X}_M) = \boldsymbol{X}_M + \boldsymbol{X}'_M, \tag{38}$$

$$\mathrm{MABA}(\boldsymbol{X}_H) = \boldsymbol{X}_H + \boldsymbol{X}'_H. \tag{39}$$

3.5 Condition Control in U-net

To fuse RGB images and Wi-Fi CSI data, we incorporate their embeddings into the denoising stage of the diffusion model. The RIE encodes RGB images to produce image embeddings, while the CCE encodes CSI data to produce CSI embeddings. Both encoders are trained simultaneously with the U-net. The MABA mechanism is working between the RIE and CCE to enhance multi-modal fusion. As shown in Fig. 2, we insert the MABA after multiple RB_2 and RB_1 residual blocks in RIE, as well as after the CTransEnc module of CCE.

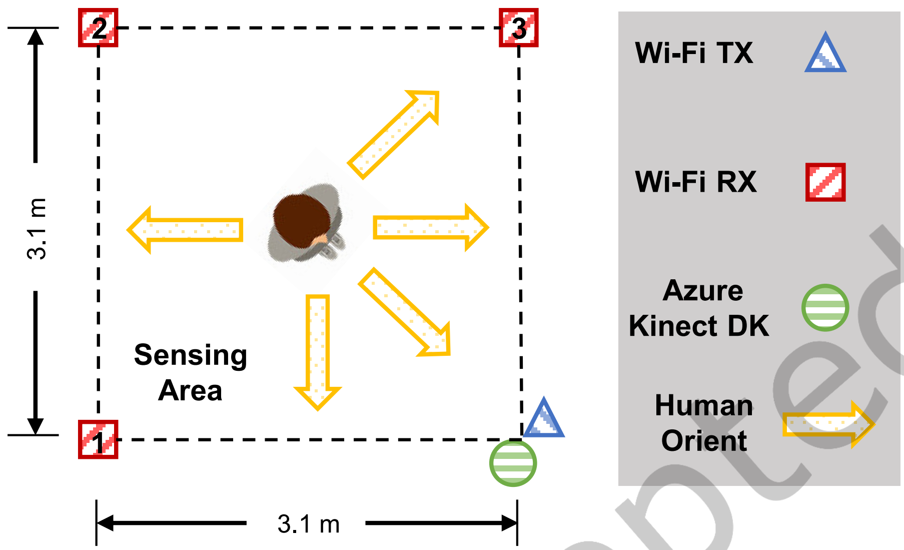


Fig. 3. Layout of Experiment Space

After the final Cross-Attention layer in Fig. 2, the feature vectors of the RGB image $\boldsymbol{X}_M$ and Wi-Fi CSI $\boldsymbol{X}_H$ are added to the intermediate feature representation within the U-net blocks as the condition to guide the process of denoising. For each channel of the intermediate representation, a broadcasting mechanism is employed to incorporate the corresponding values, guiding the noise prediction process for depth estimation. The conditional control process can be described as follows:

$$\boldsymbol{X}_{multimodal} = \boldsymbol{X}_M + \boldsymbol{X}'_H \tag{40}$$

$$\boldsymbol{N}' = \boldsymbol{N} + \boldsymbol{X}_{multi-modal}, \tag{41}$$

where $\boldsymbol{N}$ and $\boldsymbol{N}'$ are intermediate representations of the noised depth map $\boldsymbol{x}_t$ before and after conditional control within each U-net block (i.e., Down Sampling, Bottleneck, and Up Sampling in Fig. 2), and $\boldsymbol{X}_{multimodal}$ is the multimodal condition.

4 Evaluation

4.1 Methodology

Implementation. We preprocess the raw Wi-Fi CSI data using MATLAB, converting it into tensors for model training. The *WiViD* model is implemented in PyTorch, with training conducted on an NVIDIA GeForce RTX 3090. During training, the number of diffusion steps T is set to 250, and a cosine annealing strategy adjusts the learning rate from 1e-4 to 1e-7. The model is trained for 100 epochs with a batch size of 16. During testing, the number of accelerated denoising steps T_a is set to 3.

Dataset and Data Preprocessing. For our experiments, we preprocess the XRF55 [30] dataset, which includes RGB images, Wi-Fi CSI data, and depth maps as ground truth. The dataset consists of 10,000 RGB images, created by frame-sampling 400 videos, with the spatial layout shown in Fig. 3. To streamline computations, the original 1280×720 images and depth maps are resized to 160×90 pixels. The Wi-Fi CSI data for each frame are collected from three Wi-Fi RXs and concatenated into a unified dataset. The data is then shuffled and split into training and testing sets for *WiViD* training and evaluation.

Evaluation Metrics. Depth estimation effectiveness is evaluated using various metrics; we adopt two commonly used ones: error values and threshold accuracy. Lower error values correspond to better depth estimation, while higher threshold accuracy indicates improved performance.

The Absolute Relative Error (ARE) and Relative Squared Error (SRE) measure the overall discrepancy between the predicted depth and the ground truth.

• Absolute Relative Error (ARE):

$$\mathrm{ARE} = \frac{1}{N}\sum_{i=1}^{N}\frac{\left|d_{\mathrm{true},i} - d_{\mathrm{pred},i}\right|}{d_{\mathrm{true},i}}. \tag{42}$$

• Squared Relative Error (SRE):

$$\mathrm{SRE} = \frac{1}{N}\sum_{i=1}^{N}\frac{\left|d_{\mathrm{true},i} - d_{\mathrm{pred},i}\right|^2}{d_{\mathrm{true},i}}. \tag{43}$$

Root Mean Square Error (RMSE) and Root Mean Square Logarithmic Error (RMSELog) are particularly sensitive to local depth estimation inaccuracies.

• Root Mean Square Error (RMSE):

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}|d_{\mathrm{true},i} - d_{\mathrm{pred},i}|^2}. \tag{44}$$

• Root Mean Square Logarithmic Error (RMSELog):

$$\mathrm{RMSELog} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left|\log d_{\mathrm{true},i} - \log d_{\mathrm{pred},i}\right|^2}. \tag{45}$$

Threshold accuracy measures the similarity between the predicted depth and the ground truth, assessing the model's accuracy within a specific depth range. It comprises three primary metrics: δ_1, δ_2 and δ_3, each corresponding to a different depth threshold.

• Threshold accuracy (δ_k) is defined as the percentage of $d_{\mathrm{pred},i}$ where $\max\left(\frac{d_{\mathrm{pred},i}}{d_{\mathrm{true},i}}, \frac{d_{\mathrm{true},i}}{d_{\mathrm{pred},i}}\right) < 1.25^k$, for $k = 1, 2, 3$.

It is important to note that the ground truth depth map contains pixels with undefined depth, which are excluded from the calculation of the metrics.

4.2 Overall Performance

To assess the performance of *WiViD*, we select four prominent monocular depth estimation algorithms as comparison benchmarks: DORN[8], VA-DepthNet[16], VGGT[31], and Depth Anything V2[32]. DORN transforms depth prediction into an ordinal classification problem, while VA-DepthNet uses variational constraints with an encoder-decoder. VGGT employs a Transformer architecture for multi-image 3D tasks, and Depth Anything V2 enhances accuracy via synthetic training and large-scale pseudo-labeling.

As shown in Fig. 5(a) and Fig. 5(b), *WiViD* outperforms existing monucular methods across key metrics. It achieves an ARE of 0.034, surpassing DORN at 0.153, VA-DepthNet at 0.059, VGGT at 0.062, and Depth Anything V2 at 0.051. For δ_1, *WiViD* reaches 0.972, significantly higher than DORN's 0.786, VA-DepthNet's 0.943, VGGT's 0.941, and Depth Anything V2's 0.952. *WiViD* also demonstrates superiority in SRE with 0.038 compared to

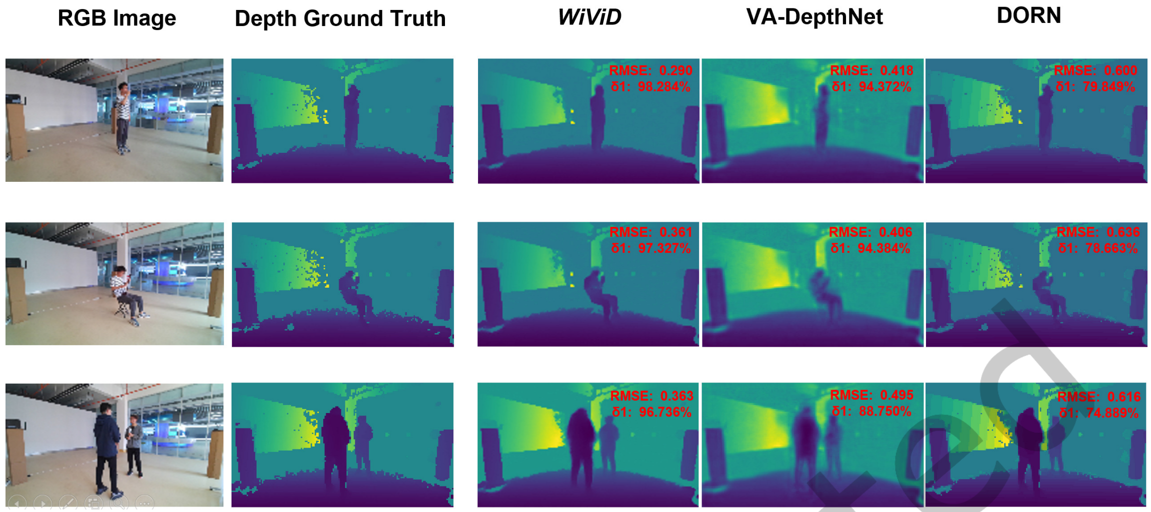


Fig. 4. Depth Estimation Samples of *WiViD* and Benchmarks

DORN's 0.097 and VA-DepthNet's 0.049, as well as in δ_2 with 0.989 versus competitors' scores up to 0.987. The generated depth map in Fig. 4 further highlights *WiViD*'s precision in capturing both static scene depth and human action-induced changes. These results confirm that *WiViD* delivers superior depth estimation accuracy compared to state-of-the-art monocular approaches and demonstrate the valuable role of Wi-Fi signals in enhancing depth estimation performance.

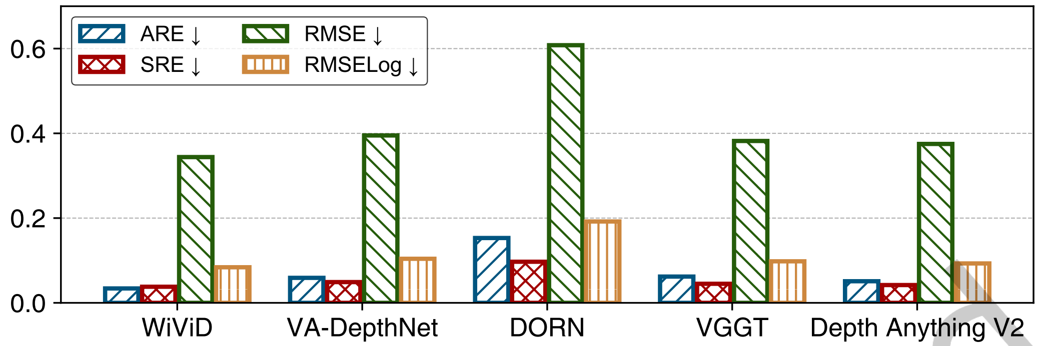
4.3 Micro Benchmarks

4.3.1 Effectiveness of Multimodal Fusion. To evaluate the effectiveness of multimodal fusion in *WiViD*, we conduct an experiment comparing the performance of three methods: using RGB images alone, Wi-Fi CSI data alone, and their combination. This comparison allows us to investigate how fusing RGB images with Wi-Fi CSI improves depth estimation. Additionally, we explore the interaction between these two modalities and their relative importance during fusion to better understand their combined effect on depth estimation performance.

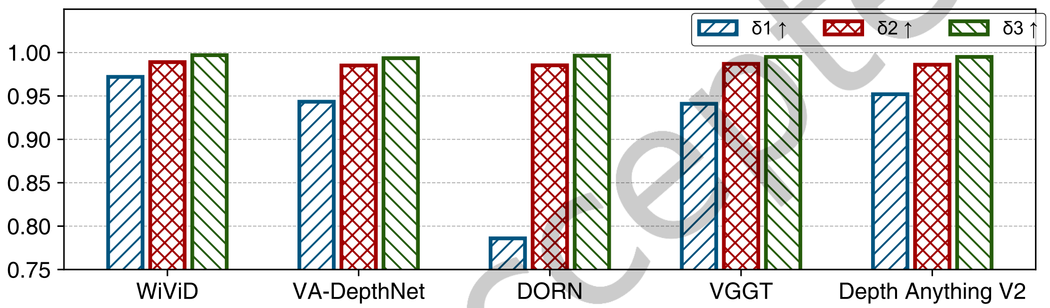
We remove one of the modalities to evaluate depth estimation results for "Vision Only" and "Wi-Fi Only". As shown in Fig. 6(a) and Fig. 6(b), the multimodal fusion approach in *WiViD* consistently outperforms both single-modality methods across all evaluation metrics. Specifically, the RMSE values for the multimodal method, "Vision Only", and "Wi-Fi Only" are 0.344, 0.380, and 0.700, respectively, while the δ_1 values are 0.972, 0.970, and 0.910. These results highlight that the multimodal fusion method improves depth estimation performance, underscoring the effectiveness of combining vision and Wi-Fi data for more accurate depth estimation.

To explore the impact of fusing Wi-Fi and vision for depth estimation, we selected two sets of samples for comparative analysis, as shown in Fig. 7. The results clearly show that the "Vision Only" method fails to accurately capture many depth details compared to the multimodal fusion approach. Additionally, the "Wi-Fi Only" method struggles to detect depth changes caused by human movements, further emphasizing the limitations of relying on a single modality. These findings highlight the significant advantages of combining Wi-Fi and vision data, as the fusion method provides more comprehensive and accurate depth estimation.

Monocular visual depth estimation extracts visual cues from a single image to infer depth, providing estimates based on the available visual information. However, this method lacks direct physical depth data, making it less accurate, particularly in complex or dynamic environments. In contrast, Wi-Fi CSI is directly tied to the



(a) Estimation Error

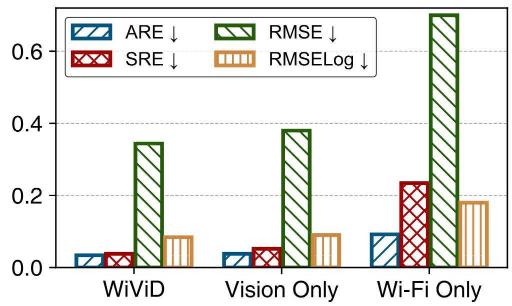


(b) Threshold Accuracy

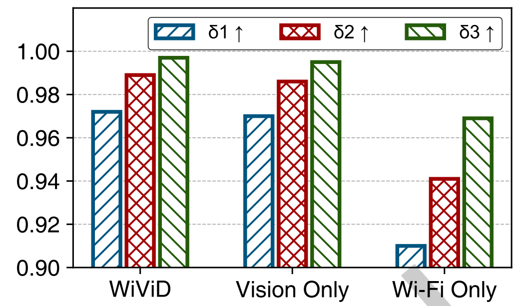
Fig. 5. Overall Performance Comparison with Benchmarks

environment's geometric structure, offering valuable physical depth data. Although Wi-Fi CSI provides deeper insight into the space's layout, it is limited when used alone due to its lack of pixel-level resolution and texture details, which are crucial for fine-scale depth estimation. Our observations show that fusing Wi-Fi CSI with RGB images allows the system to benefit from more detailed and complementary information, significantly improving depth estimation accuracy. This fusion capitalizes on the strengths of both modalities, with RGB images offering rich visual details and Wi-Fi CSI providing reliable geometric depth data. When RGB images alone are insufficient for accurate depth estimation, integrating Wi-Fi CSI enhances the system's robustness and performance, making it more effective in the real scenario.

Furthermore, to evaluate the cross-domain depth estimation capabilities of *WiViD*, we conduct a cross-person evaluation using a three-person leave-one-out protocol. Training on two people and testing on the unseen third yields three distinct splits for comparison against the vision-only baseline, as shown in Tab. 1. The results demonstrate that *WiViD* maintains robust performance without severe degradation when generalizing to unseen people, consistently yielding lower error rates across all splits. Therefore, by leveraging complementary multimodal cues, *WiViD* effectively mitigates the distribution shifts caused by cross-person variations, delivering enhanced accuracy and robustness in practical cross-human scenarios.



(a) Estimation Error



(b) Threshold Accuracy

Fig. 6. Performance Comparison between Multimodal and Single-modal

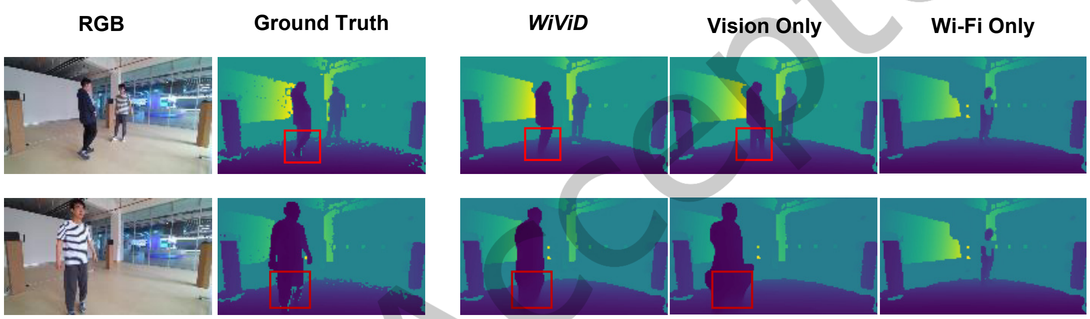
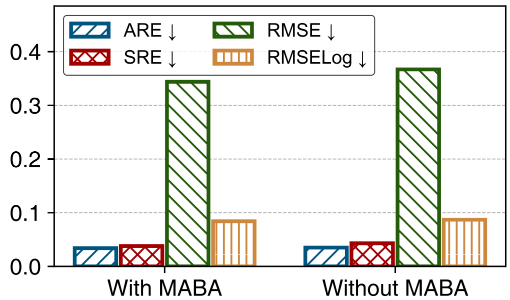


Fig. 7. Depth Estimation Samples of Multimodal and Single-modal

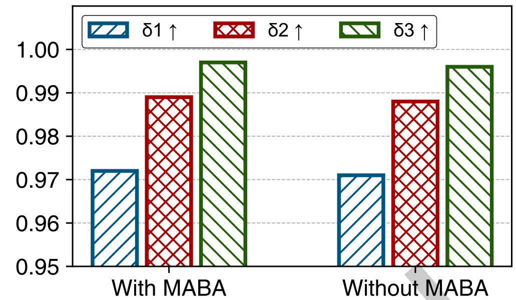
Table 1. Cross-Person Performance Comparison of Multimodal and Vision-Only

Method	***WiViD***			**Vision Only**		
Train/Test	**1&2/3**	**1&3/2**	**2&3/1**	**1&2/3**	**1&3/2**	**2&3/1**
ARE ↓	0.037	0.039	0.038	0.037	0.040	0.038
SRE ↓	0.045	0.048	0.047	0.049	0.051	0.049
RMSE ↓	0.378	0.400	0.389	0.406	0.428	0.417
RMSELog ↓	0.086	0.091	0.087	0.088	0.092	0.090
δ_1 ↑	0.967	0.962	0.965	0.964	0.960	0.962
δ_2 ↑	0.986	0.981	0.984	0.984	0.979	0.983
δ_3 ↑	0.995	0.991	0.993	0.995	0.990	0.992

4.3.2 Effectiveness of Multi-Modal Alignment Bidirectional Cross-Attention (MABA). To investigate the effectiveness of MABA, we design an experiment, specifically developing a baseline model without MABA and



(a) Estimation Error



(b) Threshold Accuracy

Fig. 8. Performance With and Without MABA

an improved model with MABA. Both models are trained and evaluated under identical conditions, and the effectiveness of MABA is assessed by comparing their performance.

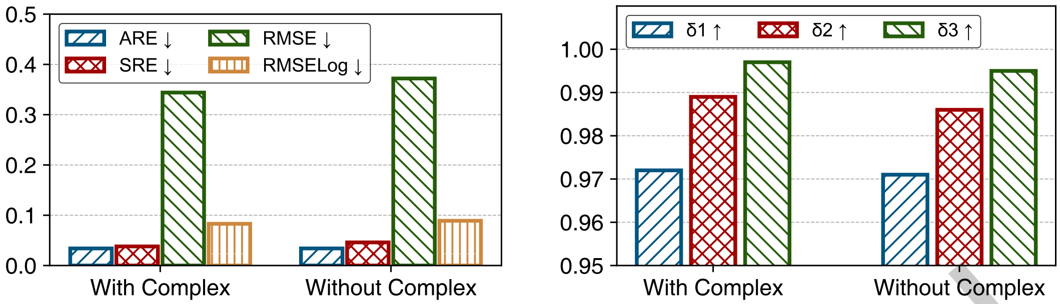
As depicted in the Fig. 8(a), regarding estimation error metrics, the model with MABA achieves an ARE of 0.034 and an SRE of 0.038, surpassing the 0.035 and 0.043 of the model without MABA. The RMSE and RMSElog are 0.344 and 0.083, outperforming their respective values of 0.367 and 0.087 in the model without MABA. For threshold accuracy metrics, shown as Fig. 8(b), δ_1 is identical at 0.972, while δ_2 and δ_3 of the model with MABA reach 0.989 and 0.997, exceeding the 0.988 and 0.996 of the model without MABA. These results confirm the effectiveness of MABA in multimodal fusion.

The effectiveness of MABA in multimodal fusion lies in its ability to better capture and utilize complementary information between different modalities. Specifically, MABA enables dynamic interaction and information exchange across modalities, allowing the model to integrate features more thoroughly from RGB images and Wi-Fi CSI. By adaptively weighting features from different modalities, MABA strengthens the model's ability to extract critical information, thereby enhancing the accuracy and robustness of depth estimation. As a result, integrating MABA into *WiViD* enhances its performance, enabling it to achieve superior results in the multimodal depth estimation task.

4.3.3 Effectiveness of Complex-Valued Neural Networks in Wi-Fi CSI Encoder. To investigate the effectiveness of complex-valued neural networks in the Complex-Valued CSI Encoder (CCE), we design an ablation experiment. We develop a baseline Wi-Fi CSI Encoder without complex-valued neural networks, compared with the one that incorporates complex-valued neural networks. Both models are trained and evaluated under the same conditions, and the performance comparison determines the effectiveness of complex-valued neural networks.

As shown in Fig. 9(a), the CSI Encoder with complex-valued neural networks achieves an SRE of 0.038, significantly lower than the 0.046 of the model without complex-valued neural networks. The model with complex-valued neural networks attains an RMSE of 0.344 and an RMSElog of 0.083, both outperforming the baseline's 0.372 and 0.089. Regarding threshold accuracy metrics in Fig. 9(b), the δ_2 and δ_3 of the model with complex-valued neural networks reach 0.989 and 0.997, respectively, higher than the baseline's 0.986 and 0.995. These results confirm that integrating complex-valued neural networks into CCE effectively reduces estimation errors and improves threshold accuracy.

The CCE model with complex-valued neural networks outperforms the real-valued counterpart because it effectively encodes the Wi-Fi CSI's amplitude and phase information, capturing subtle signal features. Notably,

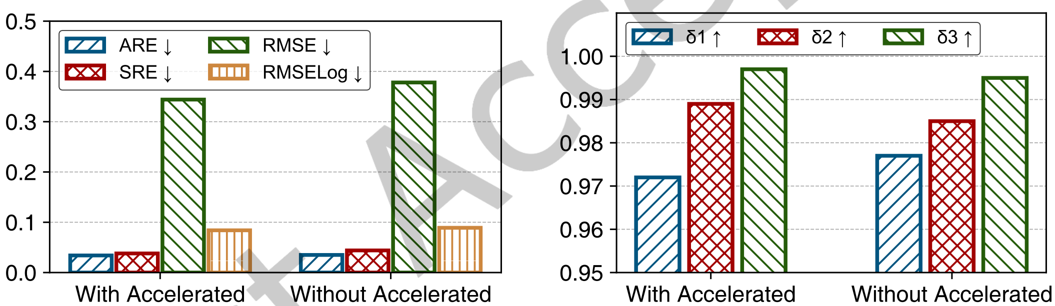


(a) Estimation Error

(b) Threshold Accuracy

Fig. 9. Performance With and Without Complex-Valued Neural Networks in Wi-Fi CSI Encoder.

while the real-valued baseline underperforms the complex-valued model, it still exceeds vision-only performance, showing that even real-valued CSI encoding adds complementary information to visual data and highlighting the value of integrating wireless signals.



(a) Estimation Error

(b) Threshold Accuracy

Fig. 10. Performance With and Without Accelerated Sampling.

4.3.4 Effectiveness of Accelerated Sampling. We design an experiment to assess the impact of accelerated sampling strategies on *WiViD* 's depth estimation performance and sampling speed. Specifically, we implement *WiViD* with and without accelerated sampling and conduct comparative tests under identical experimental conditions.

As shown in Fig. 10(a) and Fig. 10(b), *WiViD* with accelerated sampling performs roughly the same as *WiViD* without accelerated sampling across most metrics in depth estimation. Specifically, the ARE for *WiViD* with accelerated sampling is 0.034, slightly lower than 0.035 for *WiViD* without accelerated sampling, and the SRE is 0.038, also better than 0.044 for *WiViD* without accelerated sampling. For RMSE and RMSElog, *WiViD* with accelerated sampling achieves scores of 0.344 and 0.083, respectively, both superior to the 0.378 and 0.089 for *WiViD* without accelerated sampling. Additionally, in terms of threshold accuracies, *WiViD* with accelerated sampling is slightly lower than *WiViD* without accelerated sampling for δ_1 and δ_3, with values of 0.972 versus

0.977 and 0.997 versus 0.995, respectively, but outperforms on δ_2, with values of 0.989 versus 0.985. Overall, the performance of the two sampling methods in depth estimation tasks is similar, with *WiViD* using accelerated sampling showing advantages in estimation error metrics.

Table 2. Latency of *WiViD* With and Without Accelerated Sampling

Run time / *s*	**20**	**40**	**60**	**80**	**100**	**120**
With Accelerated / *ms*	182	184	183	183	182	183
Without Accelerated / *ms*	15547	15552	15569	15547	15543	15522

As shown in Tab. 2, the latency difference between *WiViD* with and without accelerated sampling is substantial. Throughout different runtime instances, *WiViD* with accelerated sampling maintains a stable latency of approximately 183 ms, whereas *WiViD* without it reaches around 15,550 ms. Leveraging accelerated sampling, the latency is reduced by approximately 98.8%, markedly enhancing *WiViD* 's sampling speed and efficiency while preserving depth estimation quality. This optimization enables the system to execute sampling tasks more efficiently without sacrificing accuracy, thereby substantially boosting overall performance.

4.3.5 Impact of Accelerated Sampling Steps. To assess the impact of accelerated sampling steps T_a on *WiViD*'s performance, we conduct this experiment. In the diffusion model, the number of sampling steps critically affects the quality of the depth map and sampling speed: a smaller T_a demands greater noise removal per step, whereas a bigger T_a increases the sampling time and process complexity. Specifically, we evaluate the effects of different T_a values (1, 2, 3, 5, 10, 20, 50, 100, 150, 200, 250) on depth estimation accuracy and inference efficiency under otherwise identical conditions.

Table 3. Performance of *WiViD* in Different Accelerated Sampling Steps.

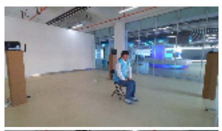
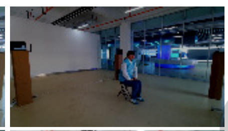
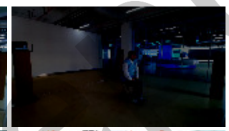
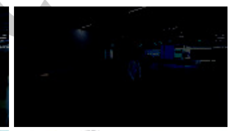
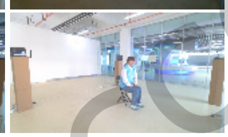
T_a	**1**	**2**	**3**	**5**	**10**	**20**	**50**	**100**	**150**	**200**	**250**
ARE ↓	1.456	0.366	0.034	0.029	0.033	0.033	0.032	0.033	0.032	0.032	0.033
SRE ↓	9.508	0.666	0.036	0.039	0.062	0.067	0.070	0.071	0.069	0.070	0.072
RMSE ↓	5.137	1.358	0.344	0.365	0.426	0.468	0.492	0.495	0.495	0.492	0.496
RMSELog ↓	1.127	0.711	0.083	0.096	0.100	0.109	0.113	0.114	0.113	0.112	0.113
δ_1 ↑	0.266	0.487	0.972	0.972	0.966	0.963	0.961	0.961	0.961	0.961	0.960
δ_2 ↑	0.389	0.698	0.989	0.988	0.981	0.977	0.975	0.975	0.975	0.975	0.975
δ_3 ↑	0.505	0.815	0.997	0.996	0.993	0.991	0.991	0.991	0.991	0.991	0.991
Latency/*ms*	64	123	182	306	625	1227	3025	6146	9110	12083	15563

The experimental results are presented in Tab. 3. As the number of accelerated denoising steps T_a increases, *WiViD* exhibits substantial performance improvements in the initial stage ($T_a = 1$ to $T_a = 3$). For example, the ARE decreases from 1.456 to 0.034, the SRE drops from 9.508 to 0.036, and the RMSE declines from 5.137 to 0.344, while the threshold accuracies δ_1, δ_2 and δ_3 also improve markedly. However, when T_a exceeds 3, the performance improvement plateaus, and some metrics exhibit slight degradation. For instance, the ARE remains nearly unchanged, from 0.034 at $T_a = 3$ to 0.033 at $T_a = 250$; the SRE slightly increases from 0.036 to 0.072; and the RMSE gradually rises from 0.344 to 0.496. These results suggest that the marginal benefit of increasing T_a diminishes, and slight degradation in some metrics may arise due to overfitting or artifact accumulation.

Meanwhile, the inference latency shows a linear relationship with the number of steps, increasing from 64 ms at $T_a = 1$ to 15563 ms at $T_a = 250$. Within the range of $T_a = 3$ to $T_a = 10$, *WiViD* achieves a good balance between performance and latency. For example, at $T_a = 3$, the ARE is 0.034 with a latency of 182ms. However, when $T_a > 50$, the latency increases significantly, while the performance improvement becomes negligible. Therefore, the experiments demonstrate that a relatively low number of steps can achieve high performance, and further increasing T_a yields diminishing performance gains but significantly increases computational costs. Selecting a moderate number of steps ($T_a = 3$) strikes an optimal balance between performance and efficiency.

4.4 Robustness Experiments

Table 4. Underexposure or overexposure for RGB image

Exposure	**0%**	**25%**	**50%**	**75%**
Underexposure				
Overexposure				

4.4.1 Robustness of Underexposure and Overexposure. To assess the impact of exposure variations on depth estimation, we simulate scenes under different levels of underexposure and overexposure. We examine three exposure levels: 25%, 50%, and 75%, as detailed in Table 4. We compare the depth estimation performance of a multimodal model *WiViD* that integrates vision and Wi-Fi data with a "Vision Only" model that relies solely on visual input. Both models degrade in performance as underexposure and overexposure intensify, leading to texture loss. However, the multimodal method *WiViD* consistently surpasses the "Vision Only" model. The performance gap is most pronounced under extreme underexposure and overexposure, where visual texture is severely degraded. The multimodal model mitigates this limitation by leveraging Wi-Fi CSI to supplement missing depth cues.

The multimodal method exhibits its greatest advantage in δ_1 and RMSE. As illustrated in Figure 11 and Figure 12, as exposure conditions deteriorate, the performance gap between *WiViD* and the "Vision Only" model widens. Under 75% overexposure, the RMSE of the "Vision Only" model increases from 0.356 to 0.399, while its δ_1 accuracy drops from 0.970 to 0.964. Likewise, under 75% underexposure, the RMSE of the "Vision Only" model increases from 0.356 to 0.395, with δ_1 accuracy dropping to 0.965. In contrast, the multimodal model consistently maintains lower RMSE values of 0.374 for overexposure and 0.385 for underexposure, while achieving higher δ_1 accuracies of 0.968 in both cases. These results underscore the multimodal model's robustness.

These findings emphasize the necessity of integrating vision and Wi-Fi data for robust depth estimation in real-world environments with dynamic lighting. By leveraging Wi-Fi CSI to compensate for missing texture information, the multimodal approach *WiViD* enhances depth estimation robustness and offers a promising solution for practical applications in visually challenging scenarios.

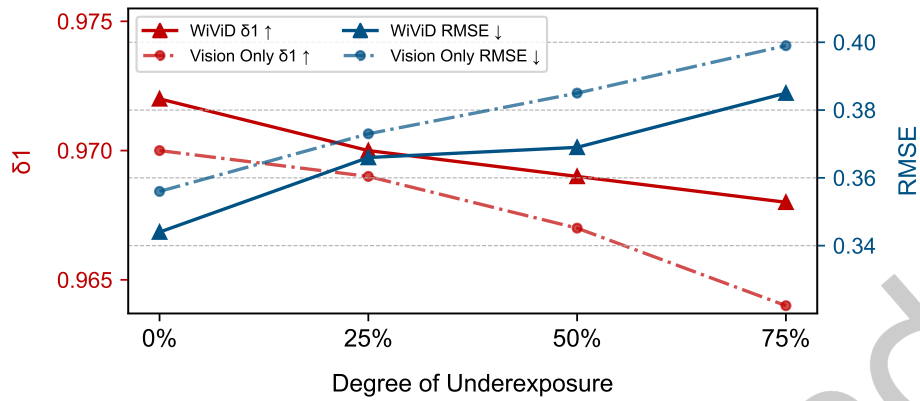


Fig. 11. δ_1 and RMSE of Underexposure

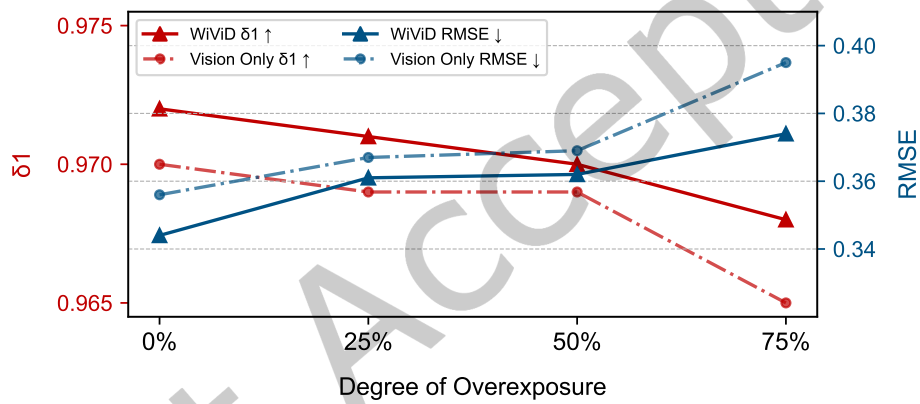
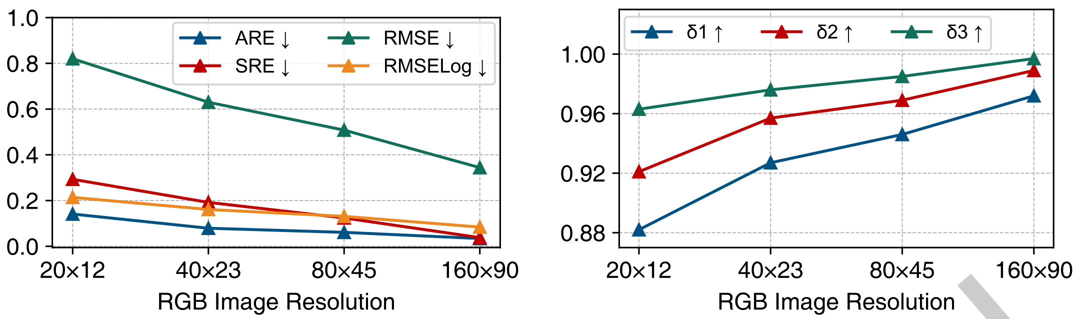


Fig. 12. δ_1 and RMSE of Overexposure

4.4.2 Robustness of RGB Image Resolution. To investigate the robustness of RGB image resolution in depth estimation performance, we design this experiment. Specifically, while maintaining a fixed output depth map resolution, we use four different RGB image resolutions: 20×12, 40×23, 80×45, and 160×90 as input for depth estimation in *WiViD*.

As shown in Fig. 13(a), all depth estimation error metrics demonstrate a significant decrease as the RGB image resolution increases. Specifically, when the resolution increases from 20×12 to 160×90, ARE decreases from 0.141 to 0.034, SRE decreases from 0.293 to 0.036, RMSE decreases from 0.820 to 0.344, and RMSELog decreases from 0.214 to 0.083. Meanwhile, as illustrated in Fig. 13(b), all threshold accuracy metrics exhibit substantial improvement. δ_1, δ_2, and δ_3 increase from 0.882, 0.921, and 0.963 to 0.972, 0.989, and 0.997, respectively.

On one hand, increasing the RGB image resolution gradually enhances the performance of *WiViD*. RGB image resolution directly affects the clarity of details and textures. High-resolution images provide richer details and sharper textures, which are essential for depth estimation. Since depth estimation performance heavily relies on the model's capacity to capture scene details, higher RGB image resolution enables more precise identification and utilization of these details, significantly enhancing depth estimation accuracy.



(a) Estimation Error

(b) Threshold Accuracy

Fig. 13. Performance of *WiViD* in Different RGB Image Resolutions

On the other hand, even at relatively low RGB image resolutions, the performance of *WiViD* remains partially robust. For RGB image resolutions of at least 40×23, δ_1 remains above 0.9, indicating that *WiViD* retains usability even at lower resolutions. This is primarily because, even at lower resolutions, RGB images preserve some level of detail. Additionally, integrating Wi-Fi CSI supplements depth information, partially mitigating the loss of visual details due to reduced resolution. This robustness allows *WiViD* to maintain depth estimation performance in resource-constrained or low-resolution scenarios, demonstrating its adaptability and reliability in real-world applications with varying camera resolutions.

4.4.3 Robustness of Wi-Fi CSI Frame Rate. To evaluate the robustness of *WiViD*'s depth estimation across different Wi-Fi CSI frame rates, we conduct this experiment. We configure Wi-Fi CSI frame rates at 10Hz, 20Hz, 50Hz, 100Hz, and 200Hz to assess *WiViD*'s depth estimation performance.

The experimental results are shown in Tab. 5, where *WiViD* demonstrates consistent and robust performance across different Wi-Fi CSI frame rates. Specifically, ARE, SRE, RMSE, and RMSELog remain stable at approximately 0.034, 0.035, 0.345, and 0.083, respectively. Meanwhile, the threshold accuracy metrics δ_1, δ_2, and δ_3 are maintained at high levels of 0.972, 0.989, and 0.997, respectively. These results indicate that *WiViD* can sustain stable depth estimation accuracy under varying Wi-Fi CSI frame rates, demonstrating its strong robustness to changes in sampling rates.

Wi-Fi CSI data demonstrates short-term stability and effectively captures scene depth information even at lower frame rates, reducing the sensitivity of depth estimation accuracy to frame rate variations. Furthermore, *WiViD* utilizes a temporal aggregation method that reduces data volume while preserving key information, thereby improving its adaptability and robustness across varying Wi-Fi CSI frame rates. Consequently, *WiViD* consistently delivers strong performance across Wi-Fi devices with different CSI frame rates, demonstrating its cross-device adaptability.

5 Related Work

Diffusion Models. Diffusion models have become pivotal in deep learning and AI. Innovations such as GLIDE [20], DALL-E 2 [22], and Imagen [26] have significantly advanced text-to-image creation by enhancing image diversity and utility. Latent Diffusion Models (LDM) [24] optimize this process by operating in a low-dimensional latent space, reducing computational demands while maintaining high-quality outputs. Additionally, in the field of radio frequency, RF-Diffusion[4] is a wireless signal generation scheme utilizing a diffusion model. These

Table 5. Performance of *WiViD* in Different Wi-Fi CSI Frame Rates

Frame rates / Hz	10	20	50	100	200
ARE $\downarrow$	0.034	0.035	0.034	0.034	0.034
SRE $\downarrow$	0.035	0.038	0.034	0.036	0.036
RMSE $\downarrow$	0.342	0.345	0.342	0.343	0.344
RMSELog $\downarrow$	0.083	0.084	0.083	0.083	0.083
$\delta_1 \uparrow$	0.972	0.971	0.972	0.972	0.972
$\delta_2 \uparrow$	0.989	0.989	0.989	0.989	0.989
$\delta_3 \uparrow$	0.997	0.997	0.997	0.997	0.997

advancements have expanded the capabilities and applications of diffusion models, making them integral to AI research and development. Unlike previous works that focused on image generation or RF signal processing, our work leverages multi-modal inputs (Wi-Fi and vision) to tackle the specific task of depth estimation. By introducing a novel cross-modal fusion mechanism, we achieve breakthrough accuracy that distinguishes our approach from existing methods.

Depth Estimation. Depth estimation is essential for various technological applications, driving the development of diverse methods to address this challenge. Convolutional Neural Networks (CNNs) are widely employed, with approaches such as those by Eigen et al. [7] utilizing multi-scale CNNs to achieve precise depth predictions. MonoDepth [10] leverages a self-supervised learning framework to estimate depth from single images, thereby eliminating the need for ground truth data and significantly enhancing efficiency. Recent advancements include monocular depth estimation using Transformer architectures, as demonstrated by MonoViT [37], which captures long-range dependencies to improve depth prediction. Collectively, these methods advance depth estimation and contribute to progress in autonomous driving, augmented reality, and robotics. Different from these methods relying on pure visual inputs, our work innovatively fuses Wi-Fi signals with visual data via a multi-modal diffusion model, enabling more robust depth estimation in complex environments.

Multi-modal Fusion Methods. multi-modal fusion is a crucial technique in deep learning that integrates information from multiple sources to enhance model performance. Simple fusion methods, such as concatenation-based approaches [17], directly merge features from different modalities at the input level, providing a unified representation but often struggling with modality-specific noise. Mid-level fusion techniques, exemplified by MulT [29], leverage cross-modal attention mechanisms to learn complex relationships between modalities, improving robustness and adaptability. Followed by decision-level aggregation, as seen in works like WiVi [40], which aligns vision and Wi-Fi in the decision module. Furthermore, diffusion-based generative fusion approaches, such as UniDiffuser [3], demonstrate the potential of probabilistic modeling in harmonizing multiple modalities. These developments have significantly advanced multi-modal learning, enhancing applications in different fields. Most previous works perform multi-modal fusion in a single stage, whereas our method implements conditional control of multi-modal fusion with each denoising step of the Diffusion model. This design allows multi-modal fusion to run through the entire iterative optimization process of depth estimation, thereby expanding the scope of traditional mid-level fusion and achieving more refined cross-modal information integration.

6 Discussion and Future Works

6.1 Strategies to Boost Wi-Fi Modality Performance

In *WiViD*, we position Wi-Fi as a complementary modality rather than an independent solution. To further leverage the potential of Wi-Fi, we propose three directions for future work. First, designing modality-specific feature enhancement modules[18] to extract more discriminative radio frequency (RF) features can alleviate the inherent resolution limitations of Wi-Fi. Second, implementing adaptive fusion weights that adjust dynamically based on environmental reliability[38], such as lighting conditions, ensures that Wi-Fi can effectively compensate for degraded visual information in real time. Third, exploring cross-modal knowledge distillation[13] from vision to Wi-Fi holds great promise for transferring depth estimation capabilities to the Wi-Fi modality, thereby enhancing its standalone performance in extreme scenarios. These strategies not only address the technical bottlenecks in RF-based sensing but also highlight the practical significance of multi-modal fusion in real-world depth estimation systems.

6.2 Dataset and Generalization

In *WiViD*, the utilization of the XRF55 dataset validated Wi-Fi-vision depth fusion, yet its limited scale and specific environmental focus restrict the evaluation of the generalization capabilities. First, enriching multi-modal dataset variability by incorporating diverse hardware configurations and environmental conditions can capture real-world heterogeneity, ensuring the model encounters the full range of real deployment scenarios. Second, integrating synthetic data generation to augment real-world samples[4], which can fill gaps in underrepresented scenarios while reducing reliance on costly physical data collection. Third, developing advanced generalization techniques, such as meta-learning[14] and domain adaptation[6], enables the model to perform well in unseen environments by learning transferable patterns that transcend specific hardware or environmental contexts. These approaches aim to address current limitations, advance Wi-Fi-vision depth fusion research, and boost *WiViD*'s reliability and applicability in real-world depth estimation tasks.

7 Conclusion

This paper proposes *WiViD*, the first Wi-Fi and vision multi-modal fusion depth estimation method based on a diffusion model with an accelerated sampling strategy. It integrates Wi-Fi and vision data using a diffusion model. The Complex-Valued CSI Encoder (CCE), Residual Image Encoder (RIE), and the Multi-Modal Alignment Bidirectional Cross-Attention (MABA) mechanism ensure the effective fusion of Wi-Fi CSI and RGB images. Experimental results demonstrate that *WiViD* surpasses existing monocular visual depth estimation methods in real-world scenarios.

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